

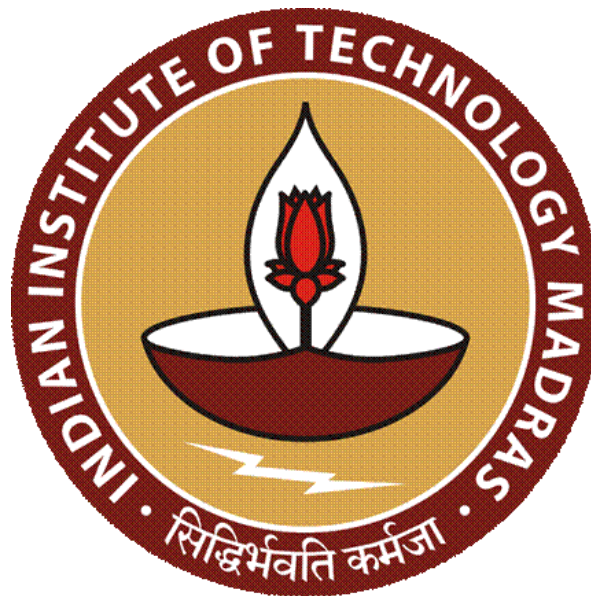
Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sale

A Final report for the BDM capstone Project

Submitted by

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CONTENTS

1. Executive Summary and Title.....	1
2. Detailed Explanation of Analysis Process/Method.....	1-3
3. Results and Findings.....	4-14
4. Interpretation of Results and Recommendation.....	15-16
4.1 Interpretation of Results	15
4.2 Recommendations	15-16

Declaration Statement

I am working on a Project titled “Leveraging Data for Smart Manufacturing: Reducing Costs & Maximizing Sticker Sales”. I extend my appreciation to Navkar Enterprises, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate

Name: Vinshi Jain

Date: 30th June 2025

1. EXECUTIVE SUMMARY AND TITLE

“NAV KAR ENTERPRISES” is a manufacturing company specializing in stickers, including customized, logo, and cartoon varieties. The organization faced significant challenges related to inventory inefficiencies, high material wastage—particularly of costly inputs like adhesive and silicone—and pricing pressures in a competitive market. These issues resulted in elevated operational costs, frequent stock imbalances, and fluctuating profitability, necessitating a data-driven approach to optimize inventory and sales management.

To address these challenges, comprehensive data was collected encompassing inventory usage, sales performance, wastage records, and competitor pricing. Descriptive statistics and analyses were conducted using tools such as Excel and Python. Key methodologies included **Pareto Analysis** to prioritize high-revenue sticker categories, **What-If Analysis** to simulate production and material usage scenarios, and trend analysis to identify demand patterns and variability across product types. These approaches provided a quantitative foundation for understanding turnover ratios, stock management gaps, and sales dynamics.

The analysis revealed that approximately 20% of sticker types contributed to 80% of **revenue**, directing focus toward optimizing these segments. Material wastage was substantial and variable, especially for adhesive and silicone, indicating inefficiencies in handling and procurement. Demand for top-selling stickers remained stable, while cartoon stickers exhibited greater volatility. These findings informed smarter procurement and inventory strategies aimed at reducing overstocking and stockouts, with an emphasis on perishable and high-cost materials.

As a result of implementing data-driven recommendations, NAV KAR ENTERPRISES observed improved inventory control, reduced material waste, and more aligned purchasing practices. This led to cost savings, enhanced operational efficiency, and better positioning against competitive pricing pressures, demonstrating the value of leveraging analytics for sustainable manufacturing growth.

2. DETAILED EXPLANATION OF ANALYSIS PROCESS/METHOD

Data Analysis Process Overview

The data analysis for NAVKAR ENTERPRISES was designed to tackle three critical business challenges: high material wastage, inefficiencies in inventory turnover and stock planning, and pressures on profitability due to competitive pricing. The project started by identifying these key issues through early data exploration and consultations with stakeholders. Following this, a meticulous process of data acquisition, cleaning, structuring, and transformation was undertaken to ensure the quality and usability of the information. Various analytical techniques, including descriptive statistics, Pareto analysis, inventory ratios, and sales trend forecasting, were applied systematically to extract meaningful and actionable insights aimed at improving operational efficiency and profitability.

Data Acquisition and Preparation

Acquiring data from NAVKAR ENTERPRISES involved overcoming initial skepticism from stakeholders who were hesitant to share proprietary information. However, through persistent communication and rapport building, valuable datasets were obtained, comprising inventory details such as quantity purchased, daily usage, remaining stock, and unit costs, along with sales data covering product types, quantities sold, revenues, and profits. Additionally, comprehensive waste data documenting item-wise wastage over time and competitor pricing information broken down by region and company were secured. Much of the raw data was initially unstructured, recorded manually in notebooks and PDFs, requiring effective transcription and cleansing using Excel and Python. The final structured datasets incorporated standardized columns like DATE, ITEM, QUANTITY, TOTAL COST, USAGE, WASTAGE, and REVENUE to provide a consistent basis for analysis.

Data Cleaning and Transformation

In order to prepare the data for rigorous analysis, key numerical fields such as TOTAL COST, QUANTITY, and REVENUE were converted from text to numeric formats to enable precise calculations. To maintain consistency across the dataset, DATE and ITEM entries were standardized for each month and product. Missing values were carefully addressed by filling them with zeros, ensuring completeness of the time series without compromising integrity. Wastage was computed as a percentage of total quantity purchased each month, allowing comparison across materials grouped into categories such as Adhesive, Silicone, Ink, and Vinyl. Competitor pricing data was reorganized by region and product type to facilitate direct comparisons and market analysis. This phase involved generating pivot tables and applying conditional formatting in Excel to highlight excessive waste, particularly any exceeding 8%, and to identify periods such as October marked by overstocking relative to actual usage.

Analysis of Material Wastage and Inventory Inefficiency

The analytical focus on material wastage and inventory inefficiency zeroed in on high-cost raw materials like Adhesive and Silicone, which showed notably elevated wastage rates. Two primary metrics were used: the inventory turnover ratio, calculated as daily usage divided by the average inventory held, and the waste percentage, derived from the proportion of quantity wasted against quantity purchased. Monthly consumption and wastage were visualized through Excel pivot tables with conditional formatting that emphasized wastage exceeding thresholds. Comparing purchase volumes to actual usage revealed overstocking tendencies, particularly in October. This analysis underscored Adhesive as the largest contributor to both waste volume and percentage, suggesting weaknesses in tracking usage accurately or inefficiencies in storage and handling practices.

Pareto Analysis for Product Prioritization

To strategically direct resources toward the most profitable segments, a Pareto analysis following the 80/20 rule was conducted on sales data. Products were ranked based on revenue contribution, with cumulative revenue percentages calculated to highlight the subset responsible for the highest proportion of income. Visualization through Pareto charts clearly showed that a small group of sticker types, likely including customized and logo designs, accounted for the majority of revenue. This enabled the business to focus efforts on these high-impact products, optimizing inventory and marketing to maximize profitability and efficiency.

Sales Trend and Forecasting Analysis

The sales trend and forecasting study evaluated volume, revenue, and price consistency across various product types and over several months. Using a combination of line charts and bar graphs, temporal patterns in sales were mapped out. It was found that customized and logo stickers exhibited stable and predictable demand, which supports steady and planned procurement cycles. In contrast, cartoon stickers showed considerable volatility in sales, demanding more flexible and responsive inventory strategies to mitigate risks of overproduction or shortage. Forecasting tools and models were leveraged to predict future sales trends more accurately, equipping the business to better align procurement and inventory with anticipated demand fluctuations, thereby improving operational agility and reducing waste.

Justification

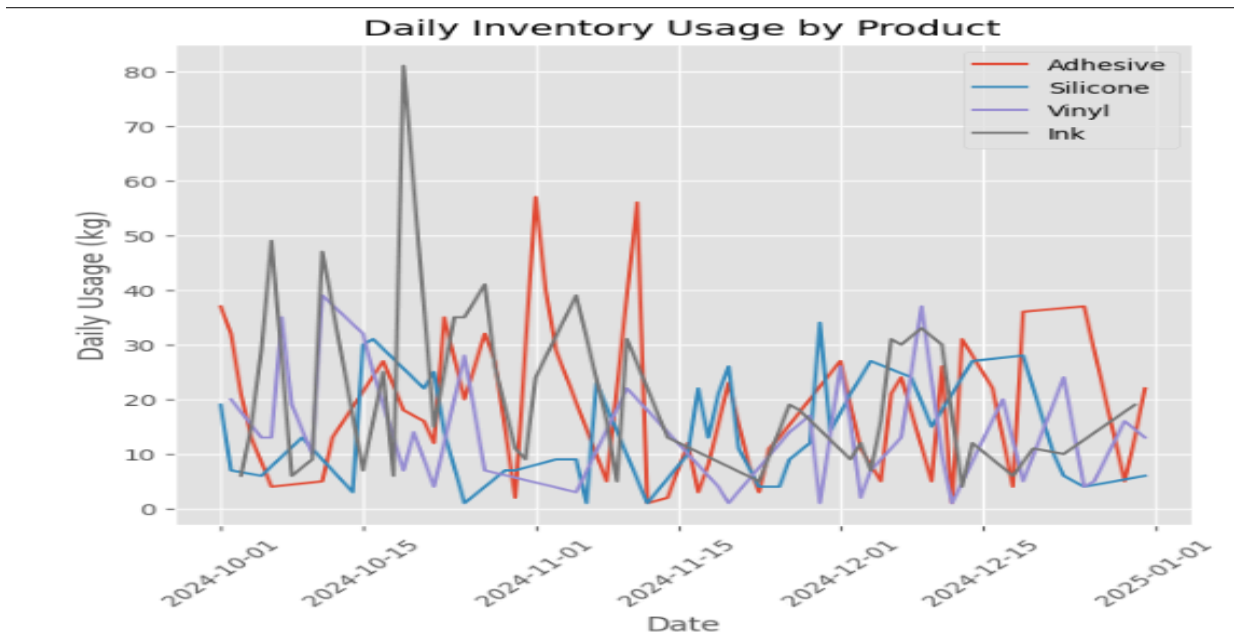
The analytical choices made for NAVKAR ENTERPRISES were carefully aligned with the core business challenges of material wastage, inventory inefficiency, and competitive pricing pressures. Diverse datasets covering inventory, sales, wastage, and competitor prices were selected to provide a comprehensive view of internal operations and market dynamics. Data cleaning and transformation ensured accuracy, enabling reliable computation of key metrics like inventory turnover and waste percentages that directly relate to stock management issues. Excel pivot tables and conditional formatting offered intuitive visualization for identifying critical

waste and overstock periods, while Python automated complex calculations efficiently. Pareto analysis focused efforts on the most revenue-generating products, linking product prioritization to profitability. Sales trend analysis and forecasting addressed demand variability, guiding adaptive procurement strategies. Forecasting with Prophet provided predictive insights to reduce stock risks and improve planning. Competitor price analysis contextualized internal decisions within market pressures, supporting smarter pricing strategies. Overall, each method and tool was purposefully chosen to generate focused, actionable insights that address NAVKAR’s specific operational challenges.

3. RESULTS AND FINDINGS:

1. Struggles with Inventory Management:

1. Daily inventory usage by products



Daily Inventory Usage Trends and Fluctuations

The displayed line graph illustrates the daily inventory usage (in kilograms) for four key raw materials—Adhesive, Silicone, Vinyl, and Ink—over the period from October 1, 2024 to December 31, 2024. Each line represents the daily consumption pattern of a specific product, offering insights into usage behavior across these materials.

High Fluctuations and Lack of Predictability

The usage patterns for all four materials are marked by high fluctuations, with frequent sharp spikes and drops. Ink, in particular, experiences the most pronounced spikes, reaching daily consumption levels above 80 kilograms around mid-October, while other materials exhibit smaller but still significant peaks. This volatility suggests that some days involve bulk consumption, potentially due to large orders or overproduction, contrasted with days of much lower usage. Additionally, there is no clear seasonality or smoothing observed in the data; usage does not follow predictable or regular patterns. This irregularity indicates potential shortcomings in demand forecasting, production scheduling, or inventory management policies.

Material-Specific Usage Patterns

Adhesive and Ink stand out with the most frequent and substantial spikes, implying either erratic demand or irregular production cycles affecting these materials. Conversely, Silicone and Vinyl show relatively steadier daily usage, albeit with occasional sudden increases; the majority of their consumption values remain clustered within lower ranges compared to Adhesive and Ink.

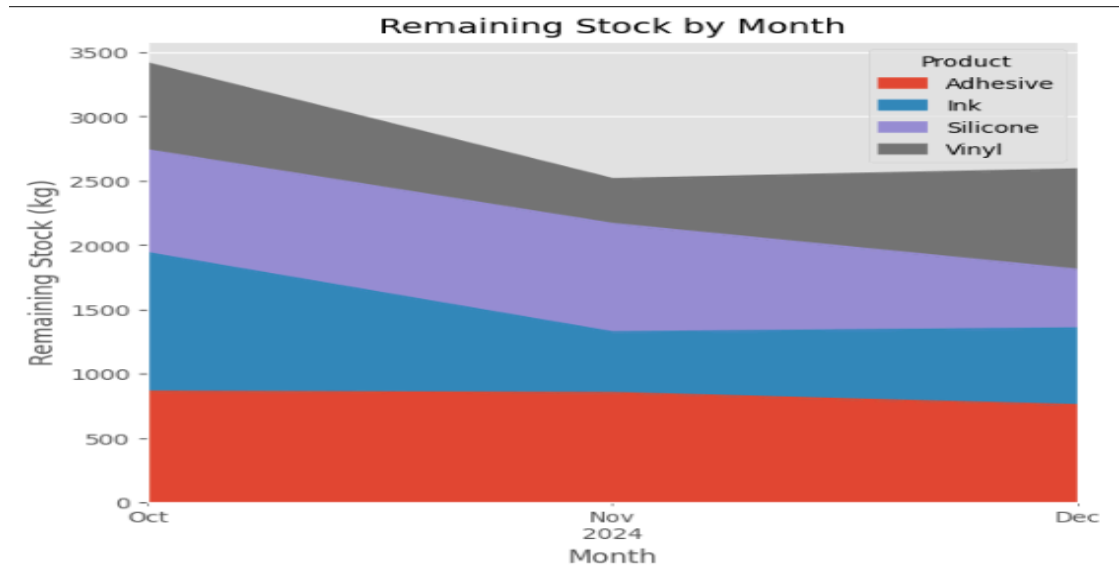
Risks Associated with Erratic Usage

The observed erratic swings in material usage create several operational challenges. For example, materials like Ink and Adhesive may be over-ordered in anticipation of peak demand, resulting in excess inventory and increased wastage during periods of low consumption. Conversely, if these demand spikes are not anticipated and planned for, stockouts may occur, potentially halting or delaying production processes.

Underlying Causes

Several factors may contribute to these patterns, including a lack of real-time stock monitoring and automated procurement or scheduling systems. Additionally, production planning may not be effectively aligned with actual demand or historical usage trends, and manual processes may result in reactive rather than proactive inventory management. Addressing these underlying causes is critical to achieving more consistent usage and reducing risks associated with inventory imbalances.

2. Remaining stock by month



Remaining Stock Trends by Month

This stacked area chart illustrates the remaining stock levels (in kilograms) of four essential raw materials—Adhesive, Ink, Silicone, and Vinyl—at the closing of each month from October to December 2024. The overall inventory shows a noticeable decline from October to November, dropping sharply from over 3,300 kilograms to approximately 2,500 kilograms. In December, this downward trend slows down, with inventory levels slightly stabilizing or increasing, indicating a potential realignment between ordering and consumption.

Material-Specific Patterns

Adhesive consistently constitutes the largest share of total inventory across all three months, though it steadily decreases, suggesting sustained and regular consumption. Ink experiences a significant drop from October to November before leveling off in December. Silicone and Vinyl exhibit similar trends, with marked declines early on followed by a plateau or slight increase in the final month.

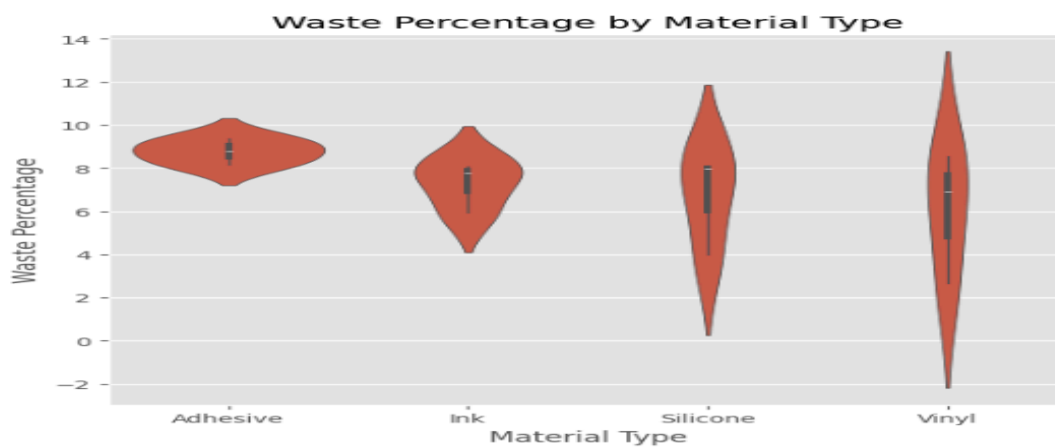
Inventory Stabilization in December

The sharp decrease in stock during November may reflect strong sales, increased usage, or an adjustment correcting possible overordering in October. The stabilization observed in December implies that procurement and consumption became more synchronized, or operational activities slowed down as the year concluded, improving inventory management during this period.

Associated Risks and Variability

The rapid depletion of stocks, especially for Ink and Silicone, underscores a potential risk of stockouts if demand remained elevated, which could disrupt production schedules. Conversely, Adhesive's relatively ample stock hints at overstocking risks, while the lower inventory levels of Ink and Vinyl pose concerns about insufficient supply, necessitating careful monitoring and inventory planning to balance these risks effectively.

3. Waste Percentage by Material Type



Waste Percentage Distribution by Material Type

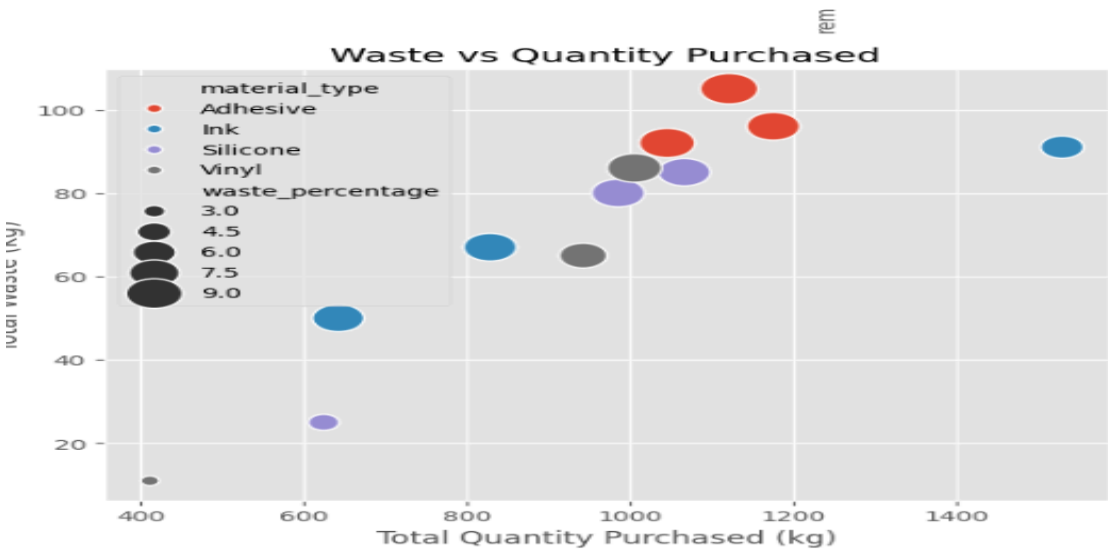
The violin plot presents the distribution of waste percentages for four key raw materials—Adhesive, Ink, Silicone, and Vinyl—providing insights into both the range and frequency of waste occurrences for each material.

For Adhesive and Ink, the waste percentages are tightly clustered within narrow ranges, approximately 8–10% for Adhesive and 7–9% for Ink, indicating consistent and repeatable waste levels. This suggests that production or storage processes for these materials are relatively stable; however, the consistently high waste percentages point to ongoing small inefficiencies that should be addressed to minimize losses.

Silicone's waste percentages exhibit a wider spread, ranging roughly from 2% up to 12%, with most values concentrated near 7–8%. The greater variability here reflects inconsistency in handling or process control, where some batches are well managed and others experience significant losses. This inconsistency signals opportunities for standardization and tighter control to reduce waste variability.

Vinyl shows the greatest range and variability in waste percentages, extending from nearly zero to approximately 13%. This substantial fluctuation indicates major inconsistencies in material usage and management. Some production periods or product batches experience minimal waste, while others face dramatic spikes likely caused by mishandling, spoilage, or excessive ordering. This highlights a strong need for improved inventory and process controls for Vinyl to achieve more predictable and reduced waste levels.

4. Waste vs Quantity Purchased



percentages fluctuating substantially—some purchase events show low waste, while others reveal inefficiency and significant loss at certain volumes.

The chart highlights critical high-waste scenarios, primarily where larger purchase quantities coincide with larger bubble sizes, particularly in Adhesive and certain Vinyl and Silicone batches. This pattern signals a risk that increasing procurement without improving stock management exacerbates material losses.

Conversely, some data points at lower purchase volumes and smaller bubble sizes—located toward the lower left of the chart—demonstrate best practices: when order quantities more closely mirror actual demand, both total waste and waste percentages remain low. This suggests that aligning purchases with realistic consumption levels is key to minimizing waste and improving inventory efficiency.

2. Rising competition and Peer Pressure:

1. Average sticker price



Average Pricing Across Sticker Categories

Cartoon stickers hold the lowest average price among the three segments, a reflection of their typically lower customization requirements and broad mass-market appeal. These stickers are

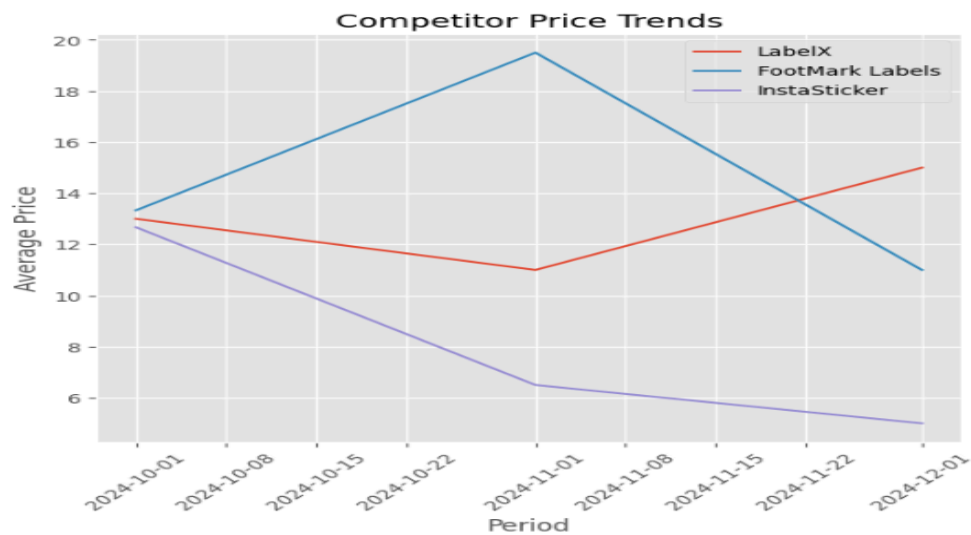
generally produced and sold in bulk, appealing to customers looking for affordable, generic designs.

In contrast, logo stickers command the highest average price. They are primarily purchased by businesses for branding and packaging purposes, often necessitating higher quality standards and strict adherence to brand guidelines. Such specialized requirements justify the premium pricing in this segment.

Custom design stickers are priced slightly below logo stickers but above cartoon stickers due to the added creative effort and personalization involved in their production. Each custom order may involve unique artwork or tailored specifications, increasing the value and cost relative to simpler cartoon stickers.

Supporting this pricing structure, market data from India indicates that cartoon stickers typically cost between ₹0.45 and ₹2 per piece when ordered in bulk as commodity items. On the other hand, logo and custom stickers can command prices well above ₹10–₹15 per piece, especially for premium or limited-run jobs, including those produced via print-on-demand or requiring bespoke designer input. This tiered pricing reflects the increasing complexity and value associated with each sticker type.

2. Competitor Price Trends



Competitor Price Trends Overview

LabelX demonstrates a relatively stable pricing strategy, with prices fluctuating between approximately 11 and 15 over the observed period. Around November 1, a slight price dip occurs

followed by a gradual upward correction. This behavior suggests a moderate, steady market approach, possibly aimed at maintaining consistent margins and avoiding aggressive price warfare.

FootMark Labels exhibits the most volatile pricing among competitors. Prices increase sharply, peaking near 20 by early November, indicative of an initial premium pricing attempt—potentially driven by heightened demand or cost factors. However, this surge is followed by a steep decline back to about 11 by December, implying customer resistance or competitive pressure forced a return to a more competitive stance.

InstaSticker follows a distinctly aggressive pricing trajectory, with a consistent price decrease from roughly 13 down to 5 across the same timeframe. This persistent downward trend characterizes classic price war tactics aimed at rapidly capturing market share, though it carries significant risks of margin erosion and potential unsustainability.

Distribution of Competitor Prices

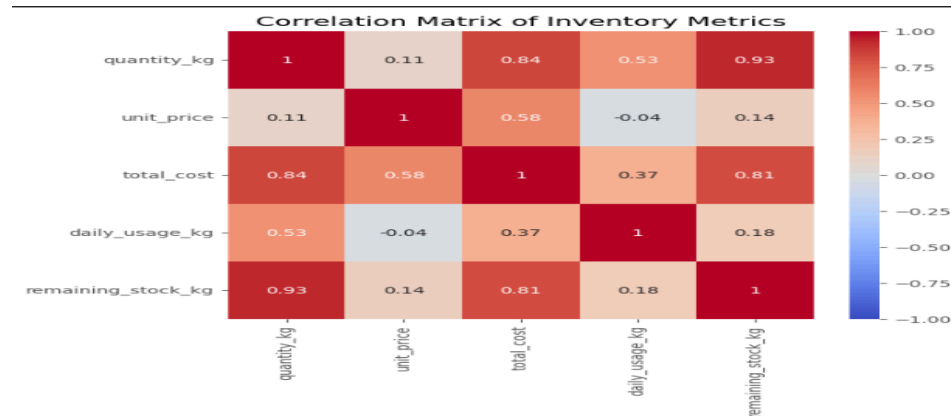
Analysis of competitor pricing reveals a multi-modal distribution, with multiple distinct peaks around price points such as 8, 10, 13-14, 16, and 20. This pattern underscores a market with no singular “going rate,” instead reflecting various pricing strategies employed by different sellers or product categories.

Competitor prices span a broad range—from as low as 4 to as high as 20—with a concentration of prices between 6 and 16. This breadth suggests a market differentiated by offerings; some competitors vie for low-end commodity or bulk orders at minimal prices, while others command premiums for specialized, higher-value, or customized sticker products. The presence of frequent price changes also indicates a dynamic competitive environment with ongoing pricing adjustments.

The absence of a dominant price point means customers likely experience diverse price offerings across suppliers, which can heighten price sensitivity and encourage frequent switching. This fragmented pricing landscape reinforces the importance of strategic pricing and differentiation for businesses seeking to balance competitiveness and profitability.

3. Lack of clarity in stock levels and Usage:

1. Correlation Matrix of inventory metrics



Correlation Analysis of Inventory Metrics

The correlation matrix reveals several strong positive relationships that highlight key aspects of NAVKAR ENTERPRISES' inventory dynamics. The near-perfect correlation of 0.93 between quantity purchased (quantity_kg) and remaining stock (remaining_stock_kg) suggests that increases in procurement directly translate to higher idle inventory. This pattern is a classic signal of overstocking and poor inventory management, where surplus purchases are not effectively consumed.

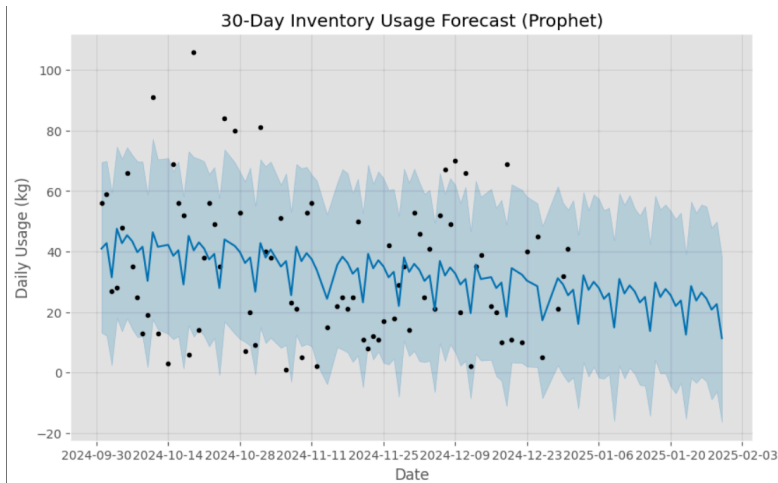
Similarly, strong correlations of 0.84 between quantity purchased and total cost, and 0.81 between remaining stock and total cost, indicate that large procurement volumes inflate both stock levels and associated costs. These relationships emphasize inefficiencies caused by purchasing decisions that are not closely tied to actual demand.

The moderate correlation of 0.53 between quantity purchased and daily usage shows that while ordering more is somewhat linked to greater consumption, usage does not rise at the same rate as procurement or stock costs. This divergence hints that increased purchasing often results in surplus stock rather than proportional usage increases.

Conversely, some variables exhibit low or negligible correlations. The nearly zero correlation (-0.04) between unit price and daily usage indicates that price changes have little to no effect on material consumption, suggesting that demand and procurement are not influenced by pricing strategies. Additionally, a weak correlation (0.18) between remaining stock and daily usage supports the notion that holding large stock does not guarantee stable or increased consumption, highlighting issues with unmonitored and fluctuating usage patterns.

Together, these findings diagnose core operational weaknesses: purchasing is not demand-driven, leading to costly overstocking and inventory inefficiencies, and usage patterns are unpredictable and poorly coordinated with procurement decisions.

2. 30 day inventory usage forecast



Inventory Usage Forecast Insights

The forecasted daily usage trend reveals a moderate downward slope, indicating an expected decrease in material consumption over the near term. This decline could be influenced by factors such as seasonal demand shifts, a slowdown in production cycles, or changes in customer requirements.

Accompanying the forecast is a broad prediction interval, represented by the shaded region, which highlights significant variability and uncertainty in daily usage. This wide range suggests that actual consumption may often differ markedly from the forecasted average, reflecting the inherently unpredictable and erratic patterns of inventory usage observed historically.

Additionally, several past observations stand out as spikes or outliers, surpassing the upper bounds of the forecast's confidence interval. These sudden surges in usage underscore the risk of unexpected demand peaks that, if not anticipated, could lead to stockouts and production disruptions. Such variability necessitates cautious inventory planning with buffers or safety stock to mitigate the impact of these demand fluctuations.

INTERPRETATION :

Integrated Interpretation of Inventory and Market Dynamics

The analysis of inventory usage and stock levels reveals considerable inconsistencies and unpredictability in both consumption and stock trends. Daily usage patterns exhibit sharp fluctuations and erratic spikes, complicating efforts to maintain optimal inventory. This instability leads to recurring issues of overstocking during low-demand periods and stockouts when unexpected demand surges occur. The monthly stock trends further underscore this disconnect, showing sharp declines followed by uneven replenishment, which indicates procurement practices are not well synchronized with actual consumption needs.

Examination of material waste and procurement patterns exposes two critical challenges. Waste percentages remain persistently high across most materials, with Vinyl and Silicone exhibiting especially wide variability. Additionally, there is a strong positive correlation between purchase volumes and waste, implying that increasing order sizes without improved inventory controls exacerbates losses rather than delivering efficiency gains. This highlights systemic inefficiencies in managing procurement and stock rotation.

From a pricing perspective, the company's internal price segmentation—where custom stickers command the highest prices followed by logo and cartoon stickers—presents clear opportunities to focus on higher-value customers. However, competitor pricing analyses reveal a fragmented landscape with no dominant market price, frequent price fluctuations, and aggressive downward adjustments by some players. Such market volatility poses the risk of destructive price wars that could erode margins if not carefully navigated through strategic pricing and differentiation.

Finally, the correlation analysis of inventory metrics confirms a strong linkage between quantities purchased, remaining stock, and total costs, but a weak relationship with actual usage, indicating that purchasing decisions are not sufficiently data-driven. The inventory usage forecast, although useful for short-term planning, also highlights the ongoing challenge of demand variability and unpredictability, necessitating integration with robust real-time monitoring and adaptive inventory control measures to effectively minimize risks and optimize stock levels.

RECOMMENDATIONS:

Problem 1: Lack of Clarity in Stock Levels and Usage

Urgent Recommendation:

Implement a real-time inventory tracking system with barcode or RFID technology within the next 3 months. This system should enable live visibility of stock movement and levels, reducing stock discrepancies by at least 30% within 6 months of deployment.

Long-Term Recommendation:

Develop automated procurement rules linked to daily usage data using forecasting tools like Prophet. Set reorder points with safety stock thresholds adjusted for usage variability. Achieve a reduction in stockouts by 40% and overstocking by 25% over the next 12 months through aligned procurement and usage monitoring.

Problem 2: Material Wastage and Inventory Inefficiency

Urgent Recommendation:

Standardize handling and storage processes for Vinyl and Silicone within 4 months, including staff training and adoption of FIFO protocols. Aim to reduce material waste percentage by 15% within 6 months post-implementation.

Long-Term Recommendation:

Establish monthly waste tracking KPIs and conduct quarterly root cause analyses for high-waste events. Target a 30% overall waste reduction across all materials within 12-18 months by improving process consistency and storage practices.

Problem 3: Competitive Pricing and Profitability Pressure

Urgent Recommendation:

Within the next 2 months, begin systematic competitor price monitoring using digital tools, tracking prices across key product segments weekly. Use this data to adjust prices dynamically while maintaining a minimum profit margin of 15%.

Long-Term Recommendation:

Develop a segmented, value-based pricing strategy tailored to distinct customer groups—prioritizing premium pricing for custom and logo stickers and competitive pricing for cartoon stickers. Target a 10% increase in overall gross margin within 12 months through effective segmentation and price differentiation.

Combined Implementation Roadmap

- 0–3 Months: Deploy real-time inventory tracking, commence competitor price monitoring, and initiate staff training on standardized handling.
- 3–6 Months: Integrate procurement with usage data using forecasting models, enforce FIFO protocols, and start waste KPI monitoring.
- 6–12 Months: Refine pricing strategy based on competitor insights, conduct root cause analyses on waste, and adjust procurement policies for safety stock and reorder points.
- 12–18 Months: Evaluate performance against KPIs—targeting reductions in waste and stock inefficiencies, improved service levels, and enhanced profit margins. Continuously optimize through data-driven decision making.

These recommendations are specific in targets, measurable via clearly defined KPIs, achievable through phased implementation, relevant to the identified business problems, and bound by realistic timelines—ensuring focused progress and sustained operational improvement.

IMPLEMENTATION:

Implementing these recommendations will significantly enhance NAVKAR ENTERPRISES' operational efficiency and profitability. Real-time inventory tracking and data-driven procurement will reduce stock imbalances, minimizing costly overstock and stockouts. Standardized processes and waste monitoring will lower material losses, improving resource utilization and reducing expenses. Dynamic pricing informed by competitor analysis will help maintain healthy margins while staying competitive in a fragmented market. Overall, these measures will lead to stronger inventory control, optimized costs, better customer satisfaction through reliable supply, and improved financial performance, positioning the company for sustainable growth.

CONCLUSION :

The combined evidence from these graphs indicates that while your business is well-positioned in a growing and competitive market, it currently suffers from inefficiencies rooted in unclear stock monitoring, reactive material ordering, unmanaged waste, and pricing pressures. However, the adoption of data-driven processes—spanning real-time stock tracking, usage-driven procurement, standardized waste controls, and strategic pricing—will transform inventory management from a liability into a strength. This will not only cut costs and prevent wastage but also allow you to compete confidently in both value and price-driven segments. By embedding analytics, discipline, and proactive controls into your operations, your business can achieve stronger margins, more reliable delivery, and a resilient foundation for growth.

Complete link with proof, data, and analysis:

<https://drive.google.com/drive/folders/1YE-5IQdkhJoblgU5RbaR9ZK40nZCVyWk?usp=sharing>